with estimate health, we take the larger of 0 or the result to accommodate for potentially negative numbers.

```
MAX((25 – (Escaped Defects * 10)), 0)
```

Using this formula, even a single escaped defect would have a big impact on the overall rating. Some example inputs and outputs are shown in table 7.5.

Table 7.5. Example inputs and outputs from the formula used to normalize escaped defects

Input	Output	Result
0	25	Perfect.
1	15	A significant impact on the total rating.
2	5	This is the highest tolerance before the number goes below 0.
3	-5	Negative numbers have a huge impact on the total rating.

Adding the elements together

To get a number from 0 to 100 with four equally important elements, the calculation should be simply adding the four numbers together. But one final part of the calculation is to create a minimum and maximum for each element. The min and max of each number should represent the absolute limits of when the team needs to take corrective action. If there are two or more escaped defects each release, the team needs to stop and figure out where the problem is. Everything running smoothly would represent the minimum measurement of each input.

To calculate the min and max, the team decided to use a scale of 0–5 for each input. This would give more weight to each input if they started to trend in the wrong direction. The final metric was calculated with the algorithm shown in the next listing.

Listing 7.2. The algorithm as Groovy code